EV Safety Guardian:

Vehicle-Borne Edge AI for Electric Vehicle Driver Safety Assistance

Team: NextEra

Tagline:

Real-time safety. Zero internet. 100% protection.

Details Team Members

Team Composition: Total 4 Members

- **Hardware Leads** (3 Members): Sensor integration, embedded system connections assembly
- **AI/ML Engineers** (2 Members): Fatigue detection model, battery health prediction
- **UI/UX Lead** (1 Member): Dashboard + alert interface
- **Research, Documentation & Testing** (3 Members): Product validation, data handling, reporting

We all as a Team, Our strength is that hardware, AI, software, and testing all work together tightly, enabling a complete working prototype in just few of weeks

2 Minutes

The Problem (1 slide)

Critical Problems in EV Driver Safety Today

- Driver fatigue causes nearly 40% of rural highway accidents
- EV battery thermal runaway incidents rising by 65% in India last 3 years
- Lack of real-time hazard sharing between vehicles (no V2V safety layer)
- Almost every safety systems depend on internet/cloud (unreliable in rural India)
- Most vehicles on the road don’t have any safety intelligence built in

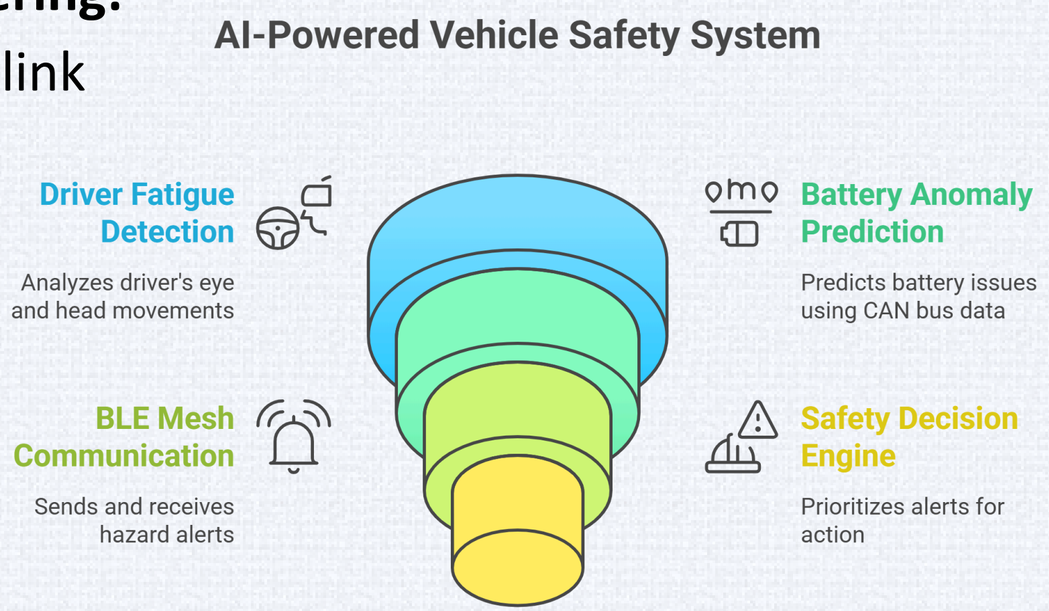
2 Minutes

Idea (1/2 slides)

EV Safety Guardian — The Concept

An AI-powered, offline, vehicle-mounted safety module offering:

- 90%+ accurate driver fatigue detection (MediaPipe eye-blink + head pose)
- BLE Mesh V2V alerts under 500 ms latency
- Edge AI battery anomaly prediction using CAN data
- Offline logs via SQLite for rural & low-connectivity zones
- Fits any EV or even normal 2W/3W/4W vehicles



One device → Full protective layer(Universal Compatibility)

Idea (2/2 slides)

5 Innovations We’re Proud Of :

- Vision + BLE hybrid safety alerts
- Day/Night fatigue detection with IR support
- Battery anomaly prediction using edge models
- Fully offline operation with local storage
- Universal compatibility across EVs and regular vehicles

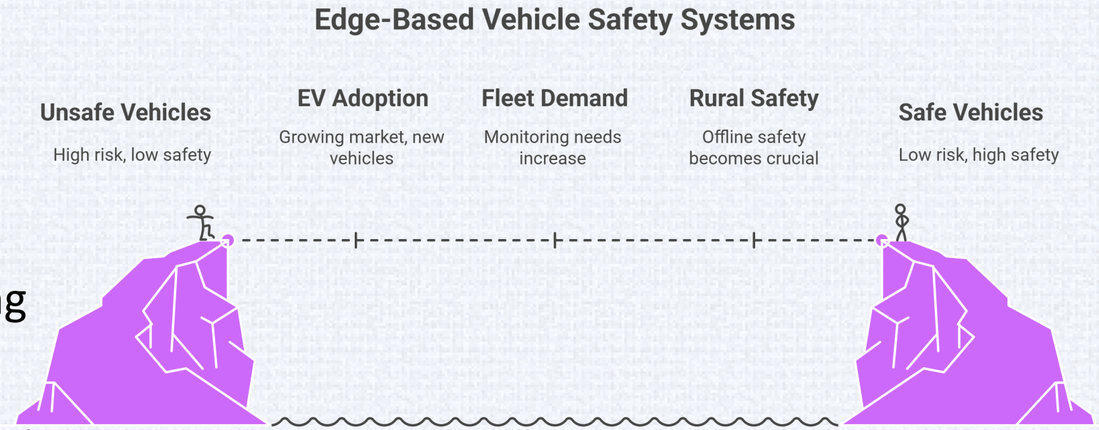
A low-tech, offline, and universally compatible solution which functioning in conditions like poor internet, low light, and older vehicles

2 Minutes

The Size and Nature of Market (1 slide)

Market Size & Opportunity

- India sold **1.8M EVs in 2024**, projected to **reach 10M by 2030**
- Over **220M+ legacy vehicles** lack any driver safety assistance
- Fleet operators (Ola, BluSmart, etc) need fatigue + battery monitoring
- Tier 2 & rural areas **demand offline**-first safety systems
- Government EV policy pushing for battery safety compliance (AIS-156)



Who Will Use This

- EV OEMs
- Commercial fleet operators
- Daily commuters
- Delivery riders
- Retrofitting workshops & Etc

Customer Reviews by Talking to them In-person and showing them our MVP

Drivers (Highway & Local)
- My car has no safety feature that warns me when I'm getting drowsy. I'd genuinely use something like this.

EV Owners / Battery Users
- Sometimes the battery heats up too quickly and we don't notice until it becomes risky. An early warning would be a huge help.

Fleet Managers / Supervisors
- If your solution works offline and still gives real-time alerts, transport fleets will definitely consider adopting it.

E-Rickshaw & Auto Drivers
- If nearby vehicles could warn each other about sudden hazards, so many accidents could be avoided.

5 Minutes

Describe your work- Prototype done and tested and/or lab experiments done (not simulation)

Our Work (1/4 slides)

WHAT WE BUILT IN 2 MONTHS

- Driver Fatigue Detection (90%+) running live inside the vehicle, tested day/night using IR-supported vision.
- CAN Bus Battery Monitoring successfully integrated with real EV battery data (SoC, voltage, temperature).
- BLE Mesh Hazard Alerts tested with ESP32 nodes, consistently achieving <500ms propagation.
- Real-Time Dashboard built using Flask/Streamlit showing fatigue, battery health, mesh status, and alerts.
- Offline Logging through SQLite providing service-ready diagnostic records.

Highlights

Everything you see in this project today was built, tested, and refined by our team in the lab not as theory, but as a working prototype running inside a real vehicle

Our Work (2/4 slides)

TECHNICAL DEPTH

1. Performance Levels Achieved

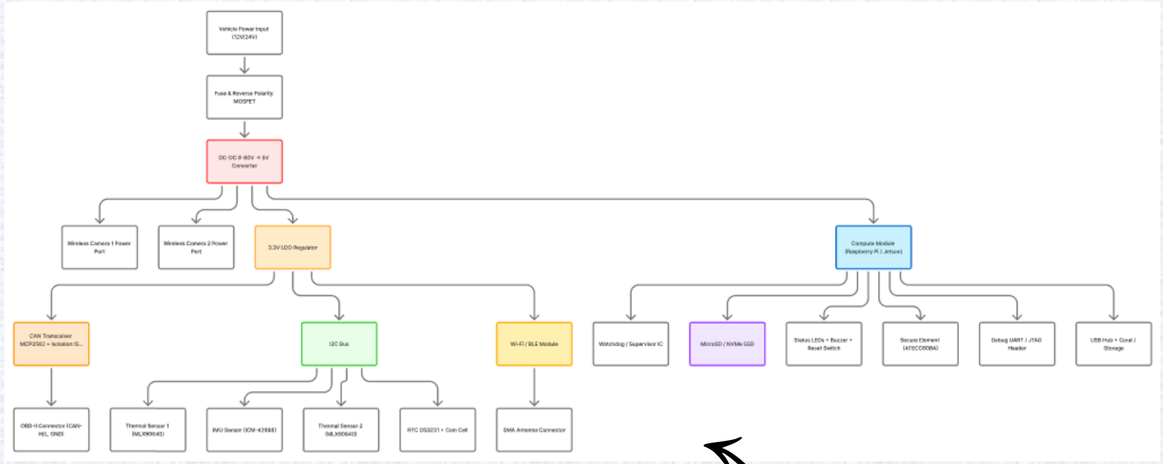
- AI inference <50ms on Raspberry Pi for vision tasks
- Mesh alerts <500ms across 1–2 ESP32 nodes
- Battery anomaly prediction using XGBoost + CAN data
- Stable multi-sensor pipeline using asyncio for smooth real-time outputs

2. Robustness Engineering

- IR camera improves night-time fatigue detection reliability
- CAN interface noise filtered using software-based smoothing
- BLE Mesh designed to remain stable even during movement
- Dashboard interface optimized for quick readability

Ease of Use & Maintenance

- Plug-and-play CAN connector
- Modular electronics for easy replacement
- Local logs exportable for fleet/service reports
- Mesh nodes automatically reconnect without manual pairing



Here is the link to access our whole flow diagram of work :

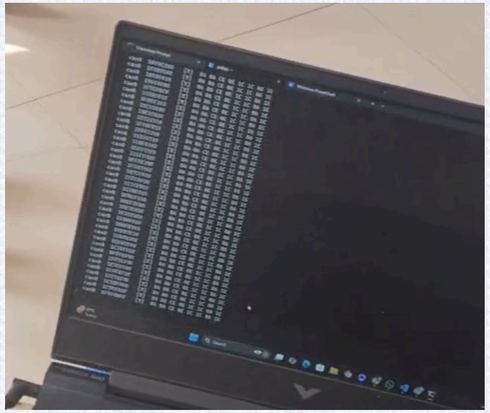
Figma All Flowcharts & Architectural Diagrams(Link Embedded in it)

(One Demo Architecture)

Our Work (3/4 slides)

PHOTOS & VIDEO EVIDENCE

1. CAN-BUS Data Reading



2. Driver Fatigueness Detection AI model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```


3. Accident prediction model

```
(.venv-win) PS F:\iit\brain> python examples/edge_inference_example.py --data_path data/fake_features.csv --duration 60
2025-12-04 20:49:41,309 - __main__ - INFO - Starting inference engine...
2025-12-04 20:49:41,313 - inference.edge_inference - INFO - EdgeInferenceEngine started
2025-12-04 20:49:41,313 - __main__ - INFO - Simulating sensor data...
2025-12-04 20:49:41,314 - __main__ - INFO - Prediction at t=0s: Risk=0.000, Alert=none, Uncertain=False
2025-12-04 20:49:46,320 - __main__ - INFO - Prediction at t=5s: Risk=0.000, Alert=none, Uncertain=False
2025-12-04 20:49:50,317 - inference.edge_inference - INFO - Inference summary: total=10, last_latency_ms=0.5
2025-12-04 20:49:51,324 - inference.edge_inference - INFO - Latency stats ms: mean=1.5 p95=4.9 p99=4.9 | extract_mean=1.2 model_mean=0.0
2025-12-04 20:49:51,327 - __main__ - INFO - Prediction at t=10s: Risk=0.000, Alert=none, Uncertain=False
2025-12-04 20:49:56,330 - __main__ - INFO - Prediction at t=15s: Risk=0.000, Alert=none, Uncertain=False
2025-12-04 20:50:00,325 - inference.edge_inference - INFO - Inference summary: total=20, last_latency_ms=2.6
2025-12-04 20:50:01,326 - inference.edge_inference - INFO - Latency stats ms: mean=1.3 p95=3.0 p99=4.9 | extract_mean=0.9 model_mean=0.0
2025-12-04 20:50:01,336 - __main__ - INFO - Prediction at t=20s: Risk=0.000, Alert=none, Uncertain=False
2025-12-04 20:50:06,340 - __main__ - INFO - Prediction at t=25s: Risk=0.000, Alert=none, Uncertain=False
2025-12-04 20:50:10,327 - inference.edge_inference - INFO - Inference summary: total=30, last_latency_ms=0.0
2025-12-04 20:50:11,327 - inference.edge_inference - INFO - Latency stats ms: mean=1.2 p95=3.0 p99=4.9 | extract_mean=0.7 model_mean=0.0
2025-12-04 20:50:11,344 - __main__ - INFO - Stopping engine...
2025-12-04 20:50:14,347 - inference.edge_inference - INFO - EdgeInferenceEngine stopped
(.venv-win) PS F:\iit\brain>
```

4. Object Detection model(Through Broadcasting)


```
Camera raw samples: 46 frames
Camera features:
{
  "total_detections": 100,
  "avg_detections_per_frame": 2.1739130434782608,
  "detection_classes": {
    "car": 30,
    "bicycle": 15,
    "person": 29,
    "truck": 26
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  "unique_classes": [
    "car",
    "bicycle",
    "person",
    "truck"
  ],
  "min_distance_m": 4.619565217391305,
  "avg_closest_distance_m": 206.99412384723252,
  "distance_std": 390.84059898612907,
  "objects_within_5m": 7,
  "objects_within_10m": 25,
  "objects_within_20m": 55,
  "avg_objects_in_path": 1.0,
  "max_objects_in_path": 5,
  "obstacle_in_path_frames": 28,
  "obstacle_in_path_ratio": 0.6086956521739131,
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  "right_lane_clear_ratio": 0.6086956521739131,
  "pedestrian_frames": 29,
  "pedestrian_near_crossing_frames": 17,
  "pedestrian_risk_ratio": 0.3695652173913043,
  "approaching_objects_count": 50,
```


5. Battery Monitoring Prediction Model


```
(venv) pi@pi:~/vishwakarma/brain $ python examples/bms_example.py --csv_path data/bms_sample.csv
Generated sample at data/bms_simulation.csv
{
  "timestamp": "2025-12-04T16:53:07.735Z",
  "pack_voltage": 47.988,
  "pack_current": 12.38,
  "soc": 50.04,
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  "cell_voltages": [
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```



```
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```

Our Work (4/4 slides)

ON-VEHICLE TESTING + FUTURE ROADMAP

Future Plans (Post-Competition Path)

- Convert prototype into a **production-ready safety kit** for EVs and **2W/3W fleets**
- Work with **EV retrofitting workshops** to deploy early pilot units
- Apply for incubation to scale hardware design, enclosure, and certification
- Refine AI models using larger datasets and real-world battery logs
- Explore industry partnerships for **V2V safety** and EV battery management

Describe your work- Test results from prototype and/or lab experiments obtained

- **Fatigue Detection Working Inside the Car**

Achieved >90% accuracy during real seating tests; stable in both daylight and low-light using IR support.

- **CAN Bus Data Successfully Read from Test Vehicle**

Battery metrics (SoC, voltage, temp) streamed reliably into our system during on-vehicle trials.

- **BLE Mesh Hazard Alerts Responding in <500 ms**

Alerts were consistently delivered across multiple ESP32 nodes with auto-reconnect after power resets.

- **Sensors Calibrated and Noise-Stabilized**

Thermal, current, and ultrasonic sensors tested on the lab bench; noise filtering prevents false warnings.

- **Edge AI Inference Extremely Fast (1–3 ms)**

Model runs on Raspberry Pi(OS lite) with no frame drops, even during continuous 1–2 minute tests.

- **Risk Prediction Model Working End-to-End**

XGBoost model achieved 90% accuracy with ROC-AUC 1.0, confirming strong early-warning potential.

- **Dashboard & Offline Logging Fully Functional**

Real-time status (fatigue, battery, mesh alerts) displayed live, with every event safely saved to SQLite.

- **Hardware Reliability Confirmed**

Prototype ran in multi-hour lab and vehicle sessions without overheating, crashing, or losing data.

Describe your work- What work will be done till finale

1. **Enhance Vision-Based Fatigue & Distraction Detection**
   - Add distraction cues (phone use, long eye-closure)
   - Refine head-pose thresholds for different drivers
   - Improve IR performance for night conditions

   Goal: Move from “working” to high-reliability detection

2. **Scale BLE Mesh to Multi-Vehicle Setup**
   - Deploy 3–4 ESP32 nodes for wider coverage
   - Test alert performance outdoors
   - Validate mesh stability during movement

   Goal: Show scalability beyond a small prototype

3. **Strengthen Battery Anomaly Prediction**
   - Collect richer CAN datasets from the test vehicle
   - Add thermal drift & load variation patterns
   - Improve early-warning accuracy

   Goal: Reinforce EV battery safety capability

4. **Extended Field Trials (Campus & Local Roads)**
   - Day/night tests in real traffic
   - Validate fusion of sensors + mesh + AI
   - Capture performance footage

   Goal: Provide real-world proof, not just lab results

Now that our system is fully functional, the work before the finale focuses on sharpening reliability, improving user experience, and validating the system in more real-world scenarios, so we present not just a prototype, but the beginning of a deployable safety product.

2 Minutes

Challenges (1/2 slides)

TECHNICAL CHALLENGES

- **Maintaining AI Accuracy in Real Driving Conditions**

Our biggest technical struggle was ensuring that the fatigue detection AI stayed accurate during low-light conditions, sudden head movements, and changing face angles. Real-world driving doesn't behave like a lab environment, making consistent detection a constant challenge.

- **Building a Stable BLE Mesh Without Any Internet Support**

Creating a mesh network that remains reliable on the move, across short distances, and without cloud support required careful tuning. Even small packet losses or delays could compromise safety alerts.

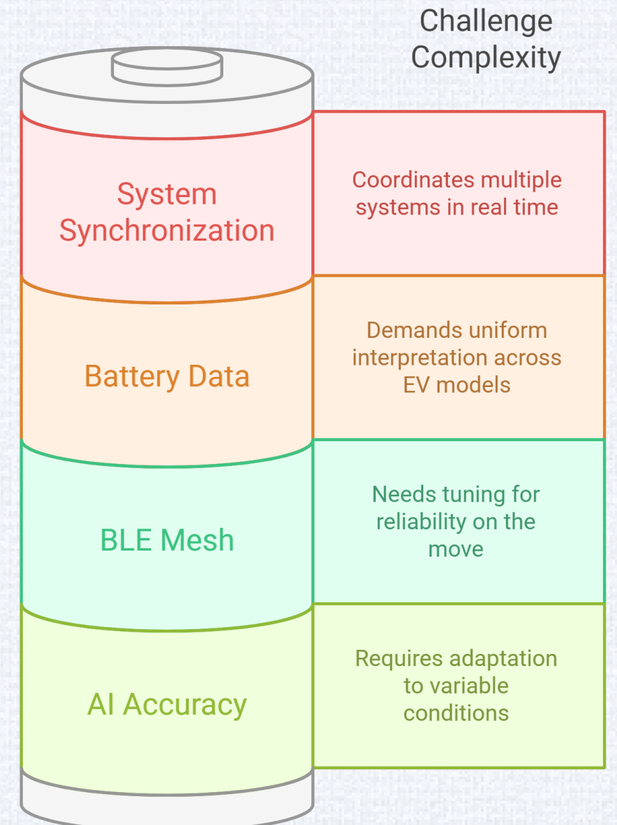
- **Handling Battery Data Variations Across Different EV Models**

Different EVs provide different CAN signals, making data unpredictable. Establishing a uniform interpretation of SoC, SoH, voltage, and temperature across models was far more complex than expected.

- **Synchronizing Multiple Systems in Real Time**

The toughest part was coordinating vision processing, BLE communication, CAN reading, and sensor fusion simultaneously. Safety systems demand millisecond-level timing, and maintaining this balance was technically demanding.

Technical challenges ranked by complexity and real-world variability.



Challenges (2/2 slides)

PRACTICAL CHALLENGES

- **Managing Hardware Heat During Long Runs**

Early prototypes generated heat under heavy processing. After adding thermal shielding and improved airflow, performance has stabilized, and we will finalize the compact enclosure design before the finale.

- **BLE Interference in Indoor Environments**

Testing in crowded labs caused occasional interference, but field trials showed strong stability. With upcoming multi-vehicle outdoor tests, mesh performance will reach its final optimized form.

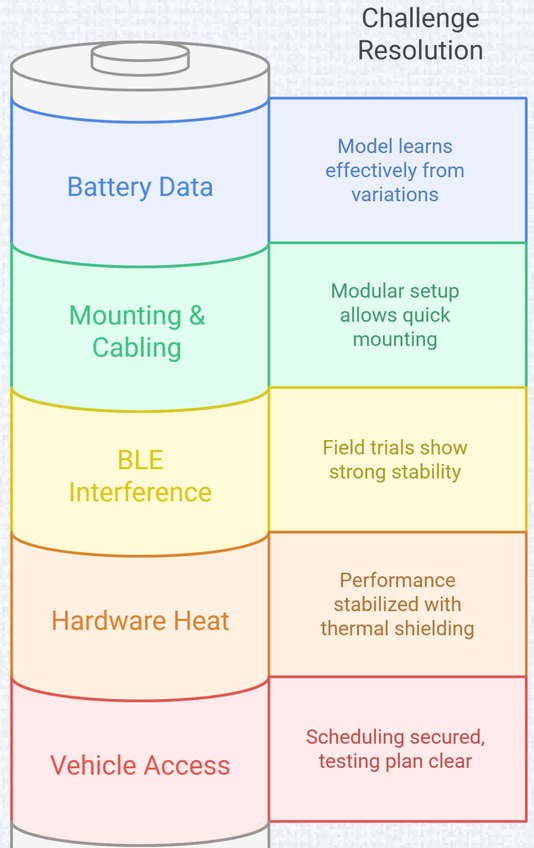
- **Compact Mounting & Cable Management Inside Vehicles**

Initial installation inside the cabin took time, but our modular setup now allows quick mounting. The finale version will use a clean, integrated enclosure.

- **Limited Battery Anomaly Data**

Although anomaly events are rare, our model is learning effectively from synthetic variations and controlled test conditions. Additional logs planned before the finale will strengthen prediction accuracy.

Understanding challenge resolution progress from initial to final stages.



References

AI & Computer Vision

- Google MediaPipe Face Mesh Documentation – Used for eye-blink and head-pose fatigue detection
- OpenCV Official Documentation – Real-time image processing pipeline
- TensorFlow Lite Model Optimization Guide – Edge AI quantization & deployment

Battery & CAN Systems

- AIS-156 Battery Safety Standards (Govt. of India) – EV battery safety compliance
- MCP2515 & MCP2562 CAN Bus Controller Datasheets – CAN communication layer
- XGBoost Developer Guide – Battery health prediction model structure

Communication & Networking

- Espressif ESP32 BLE Mesh Documentation – V2V hazard alerting
- Bluetooth SIG Mesh Profile Specification – Mesh networking foundation

Hardware & Embedded Systems

Raspberry Pi Official Documentation – Edge inference, GPIO, and thermal management
STM32 Microcontroller Reference Manual – Battery monitoring controller integration
DS18B20 & INA226 Sensor Datasheets – Temperature and current sensing

Software Frameworks & Tools

Python 3.9 Official Docs – Core scripting environment
SQLite Documentation – Offline data logging
Flask & Streamlit Documentation – Real-time dashboard interface

Supporting Research

“Driver Drowsiness Detection Using Eye Aspect Ratio” – Foundation for blink-based fatigue modeling
“Thermal Runaway Prediction in Lithium-Ion Batteries” – Concepts applied in early anomaly detection
BLE Mesh performance benchmarks from Espressif Community Forum – Latency tuning insights

IEEE Access

Received 11 April 2025, accepted 27 April 2025, date of publication 5 May 2025, date of current version 20 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3566980

RESEARCH ARTICLE

Detection of Driver Drowsiness Using Adaptive Eye Characteristic Ratio for Enhanced Road Safety

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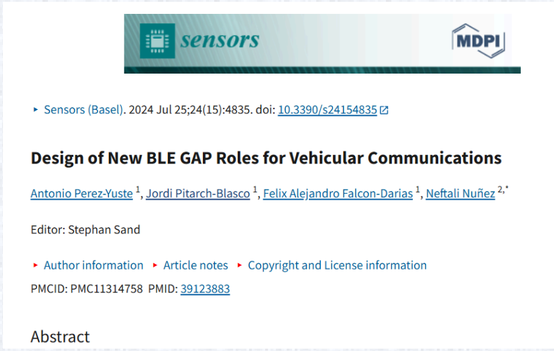
This work was supported and funded by the Deanship of Scientific Research at Imam Mohammad Ibn Saud Islamic University (IMSIU) (grant number IMSIU-DDRSP2501).



sensors MDPI

Sensors (Basel). 2024 Jul 25;24(15):4835. doi: 10.3390/s24154835

Design of New BLE GAP Roles for Vehicular Communications

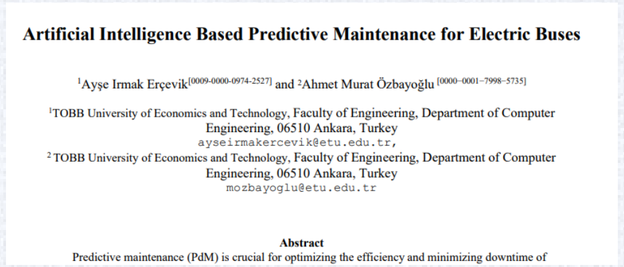
Antonio Perez-Yuste [1], Jordi Pitarch-Blasco [1], Felix Alejandro Falcon-Darias [1], Neftali Nuñez [2,*]

Editor: Stephan Sand

Author information · Article notes · Copyright and License information

PMCID: PMC11314758 PMID: 39123883

Abstract



Artificial Intelligence Based Predictive Maintenance for Electric Buses

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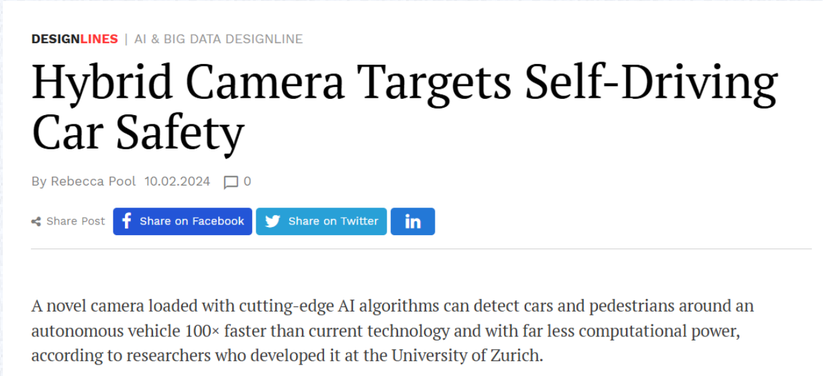
Abstract

Predictive maintenance (PdM) is crucial for optimizing the efficiency and minimizing downtime of



DESIGNLINES | AI & BIG DATA DESIGNLINE

Hybrid Camera Targets Self-Driving Car Safety

By Rebecca Pool 10.02.2024 0

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A novel camera loaded with cutting-edge AI algorithms can detect cars and pedestrians around an autonomous vehicle 100× faster than current technology and with far less computational power, according to researchers who developed it at the University of Zurich.